

Mementorium: Designing for playful and interactive learning about gender and sexuality-based marginalization

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Figure 1: Mementorium, in-game capture from a VR headset.

ABSTRACT

Mementorium is a heartfelt story about identity and belonging told through a virtual reality (VR) branching narrative. *Mementorium*'s design builds upon our previous designs and research on queer reorientations to computing and queer approaches to embodied learning in VR. The design is accomplished through the branching narrative, which supports listening to marginalized experiences and emotional co-construction of the story, and through the interaction design, which encourages play and embodied learning and centers authentic practices of belonging and becoming. This paper introduces *Mementorium*'s narrative and interaction design and research that informed the design. *Mementorium* is an approximately 30-minute single-player, room-scale interactive VR experience intended for a general audience to learn about gender and sexuality-based marginalization in science and technology.

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1 INTRODUCTION

Mementorium is a heartfelt story about identity and belonging told through a branching narrative in virtual reality. The player uncovers memories of gender and sexuality bias inside “mementos” that transport them to a dreamlike environment. They make story choices through playful interactions to transform their “mementorium” (a childhood bedroom) into a place of strength and belonging. Players can shape their story and discover, “who can we become when we belong?”

2 RESEARCH AND DESIGN

2.1 Narrative Design

A games' interactivity can challenge the idea that queer and trans representations must always show life as it is. Instead, as [Ruberg 2018] explains, games can “allow users to explore alternative ways of being and to purposefully complicate rather than distribute representations of marginalized people's lives” (p. 426). Branching narratives are a way to complicate queer and trans representations. In *Mementorium*, the narrative follows one character as they reflect on memories that shaped their sense of belonging as a queer and nonbinary person in STEM. Interaction with specific objects progresses the character's memories through time, beginning in elementary school and moving into early adulthood.

The player explores each memory by choosing a narrative branch, revealing the harm caused by marginalization. Next, the player chooses their reaction to the situation using story prompts that correspond to different emotions (e.g., anger, sadness, frustration). Having the player choose an emotional response creates an entry-point for the player to reorient themselves to marginalized experiences.

Paré et al. [2020] found that designing interactions to encourage people to reorient their attention and emotions toward marginalized experiences supports deep reflection and learning. Allowing for choices in emotional alignments with the character can help the player find themselves in the narrative they are co-creating with the character. The player's life experiences and beliefs inform their choices in how to co-construct the story. We cannot remove people's own experiences and make them feel how another person feels. However, we can ask them to make a story choice and align that action and feeling to the marginalized experience to support perspectival thinking.

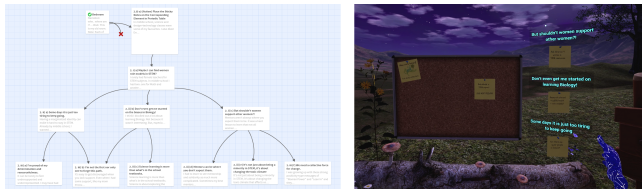


Figure 2: A narrative branch in Twine and an in-game branching interaction.

In the final stage of a narrative branch, the player must “re-story” the narrative. Re-storying is a narrative therapy technique, and its premise is that we each make sense of our lives and identities through the stories we tell and re-tell ourselves about our experiences [Denborough 2014]. Re-storying allows us to acknowledge the harms and authentically decide how to create a new narrative. The re-storying process does not erase the past harms. Instead, it offers a way to move forward. *Mementorium*'s design has players choose between individual action, interpersonal action, or social change actions. No choice is wrong. The result is that the narrative choice encourages reflection upon how our interactions shape our lives and the world around us. In this way, *Mementorium* supports learning about gender and sexuality-based marginalization and resilience as a complex phenomenon where who we can become and what the world can become are shaped through everyday interactions.

2.2 Interaction Design

In *Mementorium*, we challenge the prevailing representation of STEM as politically neutral. Such beliefs in neutrality allow for marginalized people's continued exclusion and erasure [Paré 2021]. The narrative provides one way to challenge the notion that STEM is politically neutral by sharing memories of exclusion and erasure in these environments. Our interaction design provides another way by reorienting players toward queer and trans imaginaries of STEM. Lindgren and Johnson-Glenberg [2013] suggest that to successfully leverage the affordances of virtual and embodied learning environments for education (such as mixed and virtual reality environments), the design must closely connect physical actions with the construction of new concepts. Extending beyond this focus on conceptual learning, Paré et al. [2020] found that queering embodied learning requires a critical phenomenological reorientation toward queeriness. A phenomenological approach to STEM involves an epistemological shift from conceptual learning to “a more complex

form of experience” [Sengupta et al. 2018]. Extending this approach further, a critical phenomenological reorientation “attends to experiences of recognizing normative forces and orienting oneself with counter-hegemonic and non-normative actions and people” [Paré 2021]. Furthermore, Paré et al. [2019] show that virtual reality can provide “spaces and tools that serve the purpose of re-orienting us towards dynamic, emergent experiences of gender-becoming” (p. 326). In *Mementorium*, we reorient computing through interactions that queer science and technology objects that are typically seen as politically neutral. For example, in the first interaction, the narrative reflects upon the dominant cisheteropatriarchal orientation of computing and games in elementary school. The interaction that continues into the magical realm is reminiscent of a video arcade game with a cat knight swinging a blade. A surprise is revealed at the end - a castle with queer and trans pride flags on it. Another game interaction demonstrates a way to queer programming with poetry written in coding languages, which orients us to another imagining of how we might interact with code.

3 CONCLUSION

Mementorium builds on previous design and research on queer re-orientations to computing [Paré 2021; Paré et al. 2020] and queer approaches to embodied learning in VR [Paré et al. 2019]. It accomplishes this by structuring the narrative to support player agency and reorientation toward queer and trans experiences of marginalization and resilience. It supports practices of listening and emotional co-construction of the narrative. Finally, it encourages playful interaction with queered objects that reorient science and computing toward queer and trans imaginaries that center authentic practices of belonging and becoming.

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